

Seven databases, seven problem shapes

Don't memorize features. Recognize the shape of the problem, and the service name follows.

RDS · Aurora · DynamoDB · DocumentDB · Neptune · ElastiCache · Redshift

Relational: managed SQL

AWS runs the server. You run the SQL.

RELATIONAL · THE DEFAULT

Amazon RDS

Managed MySQL, PostgreSQL, MariaDB, Oracle, SQL Server or Db2. AWS handles patches, backups and Multi-AZ failover.

- WordPress blog on MySQL
- Indie e-shop orders on PostgreSQL
- Legacy HR tool on SQL Server

Exam: managed SQL · OLTP · reduce admin work

RELATIONAL · HIGH PERFORMANCE

Amazon Aurora

AWS-native engine, MySQL/PostgreSQL compatible. Up to 5x MySQL speed, 6 copies across 3 AZs, 15 read replicas.

- Neobank: millions of transactions a day
- Multi-tenant SaaS with replica dashboards
- Super Bowl traffic spike: Serverless v2

Exam: high-perf relational · auto-scaling SQL

NoSQL: when tables don't fit

Pick by data shape: keys, documents or relationships.

KEY-VALUE

DynamoDB

Serverless NoSQL. No instance, no joins. Single-digit ms at any scale.

- Amazon.com cart
- Leaderboard, 10M players
- Dating-app swipes

Exam: NoSQL · serverless · key-value

DOCUMENT

DocumentDB

Speaks the MongoDB API. Narrow use: lift and shift a MongoDB app.

- Old CMS on MongoDB
- Catalog with flexible attributes

Exam: MongoDB compatible

GRAPH

Neptune

Nodes and edges. Relationships matter more than records.

- Friends of friends
- Bank fraud rings
- "Also watched" recos

Exam: graph · social · fraud

Speed and analytics: cache vs warehouse

One answers in under a millisecond. The other scans petabytes.

IN-MEMORY · CACHE

Amazon ElastiCache

Redis or Memcached, data held in RAM. Sits in front of your database to absorb repeated reads.

- Best-sellers computed hourly, read millions of times
- Web session store
- Top-100 game leaderboard

Exam: in-memory · sub-ms · session store

DATA WAREHOUSE · OLAP

Amazon Redshift

Columnar storage + MPP. SUM, GROUP BY and JOIN on terabytes to petabytes, in seconds.

- Marketing attribution on 200M rows
- Quarterly regulatory reporting at a bank
- Nightly product-analytics warehouse

Exam: data warehouse · BI · OLAP · petabytes

Cheat sheet: spot the shape, pick the service

The problem sounds like...	Pick
Managed SQL, OLTP, less admin work	RDS
High-performance or auto-scaling SQL	Aurora
Key-value, serverless, millions of requests/s	DynamoDB
Migrate an existing MongoDB app	DocumentDB
Relationships: social graph, fraud rings	Neptune
Overloaded DB, sessions, sub-ms reads	ElastiCache
BI dashboards, OLAP on petabytes	Redshift